Venkata Lakshmi Parimala Pasupuleti

+1-703-261-5410 | venkatalakshmiparimala@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | Portfolio | Virginia, USA

PROFESSIONAL SUMMARY

Software engineer with 8+ years building scalable, large-scale production backend systems and data processing pipelines, now specializing in generative AI. Built **AgentProbe** - an open-source LLM evaluation and grounding verification framework with a 5-metric eval engine, CI-integrated regression gate, and multi-provider LLM routing - covering correctness, grounding, latency, token efficiency, and hallucination detection across 1,240+ test cases. Built **Leasel**: an 8-agent LangGraph agentic workflow pipeline combining information retrieval (hand-rolled TF-IDF RAG), natural language processing, and grounding verification to achieve 93.6% precision on 1,240 compliance documents - every answer cited to source or escalated to a human. At Entrata, owned Python data processing pipelines and distributed systems at 100K+ MAU scale across 120+ Agile sprints. M.S. Data Analytics Engineering, George Mason University (GPA 3.87). [AgentProbe: github.com/venkatalakshmiparimala/AgentProbe]

CORE COMPETENCIES

Generative AI Systems | RAG + Grounding | Agentic Workflows (LangGraph) | LLM Evaluation Frameworks | Information Retrieval | Python · LangChain · FastAPI | ML Infrastructure · CI/CD | Scalable Distributed Systems

WORK EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern (Unpaid)

- Built an agentic workflow system orchestrated as a LangGraph stateful graph - a generative AI pipeline translating natural-language survey requests into structured flight plans, with a RAG layer (information retrieval over FAA Part 107 regulatory corpus) grounding every compliance check to cited source regulations.
- Engineered validation guardrails enforcing altitude limits, battery budgets, and geometric no-fly-zone checks before MAVLink export - deterministic logic making every final compliance decision, LLM reasoning in support.

George Mason University - College of Engineering

Aug 2025 - May 2026 · Part-time

Graduate Teaching Assistant

- Delivered technical workshops to 600+ students covering Python programming (data structures, OOP, automation) and database design (ER modeling, SQL); provided one-on-one debugging support with rubric-aligned grading and testability feedback.

George Mason University - Research

Aug 2024 - Aug 2025

Research Associate

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets across 190+ countries - $R^2 = 0.977$ regression accuracy; designed the full data processing pipeline from collection and transformation through hypothesis testing and validation.
- Applied Isolation Forest to automate data integrity checks across 3 research pipelines, surfacing anomalies that manual review missed; designed Tableau and ggplot2 dashboards improving research comprehension 40%.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 - Aug 2024 · Remote

Senior Software Engineer

- **Shipped 20+ full-stack production modules** (Python, PHP, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; led scalable microservices redesign with Docker containerization - 50% throughput improvement, 30% deployment time reduction; CI/CD pipelines eliminated 80% of manual operational workload.
- **Owned the annual HUD income-limit data processing pipeline for 7 years** - federal source rows transformed into ~180,000 domain records powering eligibility, rent, and tax-credit calculations for 100K+ housing units; every release included baseline validation, unit tests, and rollback preparation. Zero compliance violations across the ownership period.
- **Built RabbitMQ-based distributed computing system** for async gross-rent updates across thousands of tenant records - designed retry logic and idempotency guarantees eliminating double-processing failures; owned through production incidents.
- **Optimized 200+ SQL queries (30% performance gain)** through execution plan analysis, indexing, partitioning, and stored procedure refactoring across complex multi-tenant production schemas for scalable data access.
- **Led AWS S3 document migration** for Affordable Housing module - 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents across the full data processing migration.
- Mentored 5+ junior engineers; conducted 300+ peer code reviews with testability and efficiency focus; built debugging tools reducing debugging time 40%, keeping bug count below 10 for 6+ consecutive months.

Entrata India (Db Xento Systems)

Jan 2017 - Sep 2018 · Pune

Associate / Trainee Software Engineer

- Built property management modules (PHP, Python, PostgreSQL) with scalable data processing for affordable housing compliance workflows; led AWS S3 cloud migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: LLM Evaluation and Grounding Verification Framework

Open Source

2025 | Python, FastAPI, LangGraph, ChromaDB, CI/CD - ML Infrastructure

github.com/venkatalakshmiparimala/AgentProbe

Built a fully local generative AI evaluation framework: 15-endpoint FastAPI backend serving as ML infrastructure for agent evaluation, 5-metric eval engine (correctness, grounding, latency, token efficiency, hallucination rate), CI-integrated regression gate for automated testability checks before merge, and multi-provider LLM routing. Unit tests + LLM-as-judge scoring run on every push. Used to validate LeaseI across 1,240+ test cases: pass@1 81.7%, pass@3 88.3%.

LeaseI: Agentic Workflow Platform with Information Retrieval and Grounding

RAG + LangGraph

2026 | Python, LangGraph (8-agent), FastAPI, TF-IDF, natural language processing, OCR, SQLite

github.com/venkatalakshmiparimala/Lease_complaince_rag

Built an 8-agent LangGraph agentic workflow pipeline (Triage -> Retrieval -> Configuration -> Historical Incident -> Structured Data -> Reasoning -> Validation -> Human Escalation) achieving 93.6% precision and 87.9% F1 on 1,240 compliance documents. Information retrieval layer: hand-rolled TF-IDF with section-aware chunking, OCR fallback, and grounding verification - 100% citation correctness, 0% unsupported-claim rate. Every final regulated decision made by deterministic logic or a human, never by the LLM alone.

AI on FHIR: Natural Language Processing for Medical Records

2024 | Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings, NLP

github.com/venkatalakshmiparimala/AI_on_FHIR

Reduced manual clinical retrieval time ~60% by building a Flask + React natural language processing platform translating queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; enforced HIPAA-compliant RBAC across scalable structured medical records access.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering - Jawaharlal Nehru Technological University, Anantapur — 2017

SKILLS

Languages: Python, PHP, JavaScript, TypeScript, SQL, PL/SQL, R, Bash

Generative AI & Agents: Generative AI, LangChain, LangGraph, Agentic Workflows, RAG Pipelines, Grounding, Information Retrieval, Natural Language Processing (NLP), Context Engineering, Multi-Agent Orchestration, LLM Evaluation, LLM-as-Judge, Hallucination Detection, Prompt Engineering, HITL, FastAPI, ChromaDB, Vector Databases, Semantic Search, Embeddings, OpenAI API, Anthropic API

ML & Data: PyTorch, TensorFlow, XGBoost, Scikit-learn, PySpark / Apache Spark, Data Processing, NumPy, Pandas, Tableau, Power BI, spaCy, BERT, Hugging Face

Infrastructure & Systems: ML Infrastructure (FastAPI microservices, Docker, CI/CD), Distributed Computing, AWS (S3, EC2, RDS, Glue), RabbitMQ, Scalable Microservices, PostgreSQL, MySQL, Oracle, Git, Linux, Agile/Scrum